Session 4: Cleaning day

Question: What does the data look like when it is clean, and how much did we sell?

Concepts

Concept	In daily life	In Excel	The limit
DataFrame, read_csv	The whole warehouse list	A sheet, File > Open	The path must be exact
head, shape, info	First page, page count, cover	Scrolling, COUNTA	Looking first is part of the job
rename with a dict	Relabelling bins	Editing headers	Uses the dict from Session 2
filter	Keeping only the good boxes	Filter	Save the result, or it is lost. Use & with brackets
new column	Writing a value on each box	A formula column	No auto-update. Run it again
date and month	A calendar column	A date cell	Convert to a real date first
.copy()	Save As with a new name	Save As	Changing the copy cannot hurt the original
NaN	An empty box on a form	An empty cell	isna() counts them

Code patterns

```
sales = pd.read_csv("path/to/file.csv")
sales.head()
print(sales.shape)

sales_clean = sales_clean[sales_clean["quantity"] > 0]
sales_clean = sales_clean[~sales_clean["invoice"].str.startswith("C")]

both = sales_clean[(sales_clean["quantity"] > 0) & (sales_clean["price"] > 0)]

sales_clean["revenue"] = sales_clean["quantity"] * sales_clean["price"]
sales_clean["month"] = sales_clean["invoice_date"].dt.to_period("M")
```

Three common errors

Error	What to do
FileNotFoundError	The path is wrong. Copy the path from the Drive folder and check the spelling.
KeyError after rename	You used the old column name. Use the new short names.
`and` instead of `&`	Use & between conditions and put each condition in brackets.

Exercises and expected answers

Exercise	Expected
Rows after drop_duplicates	104360
After removing C invoices	102471
After quantity above 0, price above 0, real products	102257, 101933, 101459
After removing invoice 541431	(101458, 10)
Germany rows	(1514, 10)

Cleaning checklist (write the row count after each step)

Step	Rows left
Start	105,335
Drop duplicates	
Remove cancelled invoices	
Quantity above 0	
Price above 0	
Real products only	
Remove the giant order	